



BLOCK 4
EMERGING ISSUES

BLOCK 4: EMERGING ISSUES

Block 4 on Emerging Issues consists of three units.

Unit 13, “Big Data and Marketing Research,” discusses the various aspects of big data and its applications in marketing research. Types of big data that include structured data, unstructured data, semi-structured data, and quasi-structured data. Big data is characterized by six Vs: volume, variety, velocity, veracity, value, and variability. Concerns about heterogeneity, experimental variance, and statistical biases in big data are addressed for flexible and reliable research designs.

Unit 14, “Internet-based Marketing Research,” explains the purpose, applications, advantages, techniques, and trends of internet-based marketing research, including how to use existing sources, online surveys, chat-based sessions, online focus groups, social media listening, etc.

Unit 15, “Marketing Research and Social Media,” discusses the various aspects of social media from a marketing research point of view. Organizations are constantly leveraging various marketing research techniques by using various social media platforms, which are efficient and cost-effective as well. The role and benefits of social media in marketing research are illustrated by using a case study example of social media research insight.

THE PEOPLE'S
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UNIT 13 BIG DATA AND MARKETING RESEARCH

Learning Outcomes

After going through this unit, you should be able to:

- appreciate the role of big data in marketing research.
- describe the types of big data.
- explain the characteristics of big data.
- discuss the distinct advantages of big data in marketing research.
- assess the challenges associated with application of big data.

Structure:

- 13.1 Introduction
- 13.2 What Is Big Data
- 13.3 Types of Big Data
- 13.4 Characteristics of Big Data
- 13.5 Big Data and Marketing Research
- 13.6 Gaining Insights from Big Data
- 13.7 Application of Big Data in Marketing Research
- 13.8 Challenges in Application of Big Data
- 13.9 Summary
- 13.10 Key Words
- 13.11 Self-Assessment Questions
- 13.12 Project Question
- 13.13 Further Readings

13.1 INTRODUCTION

As we have seen in earlier units, data collection and analysis are important steps in the marketing research process. Marketers collect and maintain customer information, and this database is used for the formulation of marketing plans and strategies. However, as internet and communication technologies have advanced, the availability and methods of data generation have significantly changed. The use of social media and mobile technologies results in the generation of significant amounts of unstructured data, including unformatted textual, graphical, audio, and video data. Additionally, a significant amount of semi-structured data is being generated at a significant rate due to the phenomenal expansion in the digital eco-system of organizations. As a result, data is available in a very large quantity and a variety of formats, and this is where the idea of big data

is introduced. In this unit, we will discuss the various aspects of big data and its applications in marketing research to support decision-makers in making well-informed decisions.

13.2 WHAT IS BIG DATA

When we search for something on search engines like Google, Yahoo, and Bing, they know what we have searched for. Similarly, every time we access a computer, use a phone, or open a mobile app, we leave a digital footprint and add to the generation of data. So, now you can imagine how much data is being generated. An estimated 2.5 quintillion bytes (2.5×10^{18}) of data are produced every day. This rate of data creation will increase as the IoT (Internet of Things) gains popularity.

Customer databases, transaction processing systems, documents, emails, internet clickstream logs, mobile apps, and social networks are just a few examples of the many sources of big data. Additionally, it contains data that is produced by machines, including log files from servers and networks, as well as data collected from sensors on machines used for manufacturing, business, and the internet of things. Along with data from internal systems, big data settings usually include external data on consumers, financial markets, and other topics. Along with data from internal systems, big data settings usually include external data on consumers, financial markets, and other topics of interest. Big data can also include audio, video, and image files, and many big data applications use streaming data, which is analyzed and gathered continuously.

Big data can be understood same as data but it is present in very huge amount and number. It consists of the information provided by both humans and computer. Big data is so large and complex in its size and storage that it is unable to be decoded and understood efficiently by the traditional data management tools. In simple words, big data can be defined as **data that has a wider variety, arrives at a faster rate, and arrives in higher quantities**. Such data can be collected through various sources such as blogs, comments, app permissions, online surveys, virtual check-ins on e-commerce portals etc. Big data is often stored in computer databases and it is analyzed using specialized software.

13.3 TYPES OF BIG DATA

Big data can be divided on different types. Let us explain it.

Structured Data

Structural data is defined as data that can be collected, saved, retrieved, and processed in a specific format. It is simpler to organize and analyze structured data since it has some predetermined characteristics. Since each field is distinct, data from other fields can also be retrieved separately from it. Numbers and letters that are exactly aligned with the rows and columns of tables make up structured data. Data in an Excel sheet that is marked in a consistent manner and is simple to sort, update, edit, and extract is one example of structured data. Additional examples include sales records, train or airplane reservation systems, transactions, financial data, etc. Due to the data's same format, organizations derive the most value from it. Because of this, structured data is very beneficial.

Unstructured Data

Unstructured data lacks any kind of established hierarchy. The majority of big data comes from it, and it is also challenging to interpret. Google search results are an example of what unstructured data can look like. Unstructured data can take many different forms, such as basic text files, photos, videos, audio files, and other types of data. Photos we upload on Facebook or Instagram and videos that we watch on YouTube or any other platform contribute to the increasing of unstructured data. There isn't any form of recognized hierarchy with unstructured data. It is where the majority of big data originates, yet it is also difficult to interpret. A sample of unstructured data is the output of a Google search. Basic text files, images, videos, audio files, and other types of data are all examples of unstructured data. The photos we upload on Facebook or Instagram and videos that we watch on YouTube or any other platform contribute to the increase of unstructured data. For comprehensive sources of unstructured data, please click on the provided link <https://marutitech.com/big-data-analysis-structured-unstructured-data/>

Semi-structured Data

Semi-structured data is a combination of both structured and unstructured data. Examples of semi-structured data include web server logs, streaming data from sensors, Web application data, etc. Although preprocessed, semi-structured data does not appear to be a "typical" SQL database. Some data cannot have a predefined structure, whereas other data is collected in an organized fashion.

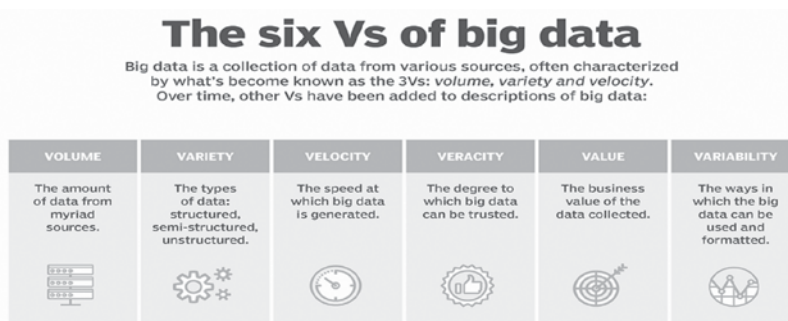
Quasi-structured Data

It falls between semi-structured data and unstructured data. Textual content with irregular data formats, such as details on which web pages a user visited first and in what sequence, serves as an example.

13.4 CHARACTERISTICS OF BIG DATA

Six Vs are frequently used to Characterize big data. They help us understand the characteristics of big data better.

Figure 13.1 Six Vs of Big Data



Source: www.techtarget.com/

Volume

The most frequently mentioned characteristic of big data is volume. The volume of data inflow is rapidly increasing in businesses. As the name advocates, the term Big Data itself is related to large and massive data size.

Stream processing systems, clickstreams, and system logs are a few examples of the sources that frequently generate huge amounts of data. Processing huge amounts of low-density, unstructured data is a necessity when working with big data. This can be unvalued data from sources like Twitter data feeds, clickstreams from websites or mobile apps, or devices with sensor support. For some firms, this data volume may approach tens of gigabytes. Several hundred petabytes might be for others. The size of data plays an important role in determining value of data. Larger is the data size, greater can be required information extracted out of it.

Velocity

Apart from huge size of data, data speed also matters. Velocity refers to the speed of data generation how fast this data is generated and processed to meet customer demands and requirements in real times. In contrast to being written to disc, the highest velocity of data often streams straight into memory. Some internet-enabled smart goods function in real time or almost real time, necessitating real-time analysis and decision-making. As a result of the constant flow of data, big data must be processed quickly in a stream-like manner. For instance, in just 30 minutes of flight, a single jet engine produces more than 10 terabytes of data. Consider the volume of information you would need to compile to study just one small aero company. Data never stops growing, so every day brings with it more information to process than the day before. This is why working with big data is so challenging.

Variety

Big data is generally heterogeneous. For instance, an enterprise's data consists of its emails, documentation, support requests, pictures, transaction records, etc. Variety refers to the diversity of sources of data, both structured and unstructured. This kind of unstructured data has implications for data mining, storage, and analysis. You must first classify and organize the data in order to gain any insights from it. Big data systems may require the storage and management of a variety of data types. Furthermore, big data applications frequently involve multiple data sets that may not be merged at first. For example, a big data analytics project might make an effort to estimate product sales by comparing data from previous sales, returns, online reviews, and customer support calls.

Value

The meaning you derive from data using specialized technologies must be useful and serve a certain purpose, such as enhancing customer experience or boosting sales. Data that can be utilized to study consumer behavior, for instance, is essential for your business since you can use the findings to create customized promotions.

Veracity

Veracity is a term used to describe how accurate and reliable a set of data is. Data quality problems resulting from the use of raw data from diverse sources might be challenging to identify. Bad data causes analysis errors that can reduce the value of business analytics initiatives if they are not rectified through data cleansing procedures. The availability of sufficient accurate data is another concern for data management and analytics in order to get reliable outcomes.

Variability

Variability is a term used to explain how quickly and how much the data under inquiry is changing. Big data sets might have various meanings or be organized differently in different data sources, which adds another level of complexity to big data administration and analytics. This parameter is crucial since even slight variations in the data can have an impact on the outcomes. It needs to regularly reevaluate your results if the variability is considerable.

Not all of the data that is collected has actual commercial worth or advantages. Therefore, before incorporating data into big data analytics projects, firms must ensure that information is related to pertinent business challenges.

Activity 1

Discuss your understanding of big data.

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13.5 BIG DATA AND MARKETING RESEARCH

As we discussed the stages of the marketing research process in unit 1, we learnt that the purpose of marketing research has been to identify research problems from decision problems and, accordingly, collect data and analyze it into information and insights that are useful to decision makers. However, big data has made it more challenging for marketers to recognize and assess markets. Big data settings have an effect on how marketing research is conceptualized, necessitating the adoption of a larger perspective as big data analytics become more common. In this section first we will discuss the key differences of big data analytics and marketing research then we will explain the process of extracting insights from big data.

You may be able to forecast Mr. Sanjeev's future behavior based on the fact that he has been getting his groceries delivered on Wednesdays between 4 and 6 p.m. for the past 3 years, as shown by your big data analytics. You might even target him with offers based on the items he frequently purchases by using the knowledge of what's on his shopping list. However, big data analytics cannot tell you why a customer shops in a particular manner. Because people and their environments change through both time and space, the predictions we can make using big data do not take unusual behavior into consideration. Again, using Mr. Sanjeev's example, you would never understand why he chose to shop one week on Friday morning as opposed to Wednesday afternoon, or why he is not shopping for a few weeks or months. Market research can help with that. Marketing research comes to the rescue in this situation.

KEY DIFFERENCES

1. The most significant difference is the volume of the data we have to process. As we have already covered, the term "big data" refers to enormous amounts of complex, unstructured, and structured data that

are only increasing in size as a result of the rapidly advancing internet and communication technologies. Information about our behaviors, such as what we buy online, share, or comment on social media, the websites we visit, the emails we receive, etc., can now be easily gathered thanks to the use of technology. Actual action is captured in real time. The amount of data collected in marketing research studies may be substantial due to the use of various techniques or sizable sample sizes, but it will never reach the same level as big data.

2. Big data offers a multitude of information, making it challenging to analyze and comprehend the data using conventional market research methods. Big data processing is one of the most difficult problems it encounters since sorting through the unlimited amounts of data to find the information you need to take too much time. You'll need specialized tools and methods to get the most out of the customer data you've acquired; big data technology and software try to control the volume, variety, and velocity of information and let you distill it down to usable insights. You also need to be careful to prevent confirmation bias in order to obtain the most value out of your data. In other words, don't just hunt for evidence that agrees with your viewpoint; also make sure your team is composed of scientists or analysts that are competent in navigating the data collection and doing a thorough analysis of it.

When it comes to handling the data, marketing research has similar problems, but on a far different level. The capability to scale the results to accurately reflect the entire market is one of the main issues faced by marketing research. For market research to be able to give the voice of the customer, obtaining representative samples and high response rates is crucial. Data would be worthless without this. A sampling tool can help you make sure your sample is big enough to be representative of your target audience. However, you should also carefully screen participants to ensure that the data you collect reflects the opinions of the majority.

3. Big data offers the ability to analyze and track behaviors in real-time at a scale that is vast enough to allow you to securely make judgments and forecast future behaviors. Big data can give businesses trends to illustrate purchasing habits, for example, over longer periods of time. But with big data, it might be simple to lose sight of the consumer as a unique individual in the mass of information. Our actions as humans are not always logical. Big data lacks the capacity to comprehend why a customer purchases a certain product and pinpoint the cause of an anomaly in predicted behavior.

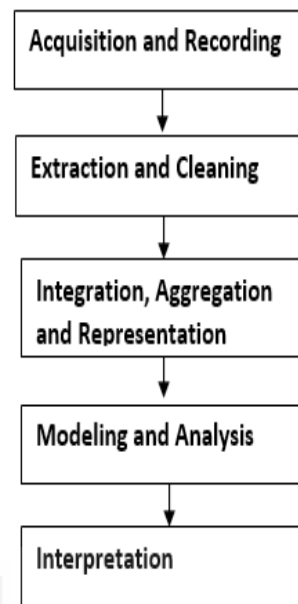
While in marketing research, participants are often asked to recall or project their behavior in a certain situation. Market research is the pursuit of meaning that takes both logic and emotion into account. It explains the reasons behind big data trends and provides information on changes. In our hypothetical case, marketing research would seek to understand Mr. Sanjeev's purchasing behaviors and the factors that influence them. As a result, predictions about his future behavior can be made based on his beliefs and attitudes.

4. While sentiment analysis on tweets or Facebook posts or data extraction from a large customer database and producing the impressive report by using various data mining techniques can be utilized to gain a general understanding of what is being said about a brand or company, a targeted marketing objective cannot be properly addressed using simply such data harvesting techniques. The targeted approaches, such as surveys, can address the more narrowly focused marketing problems. For instance, marketing research studies concentrate on a certain marketing issue, like what positioning strategies will be beneficial for the target segments. What pricing strategy is appropriate for a product that is on the decline? To know the effectiveness of advertising, to measure customer satisfaction.

Thus, we may conclude that the most obvious distinction between big data and market research is that the scale of the data determines how we handle it, how we analyze it, the insights it generates, and the specificity of goal. What you want to know about your consumers' behavior will determine which data you utilize; if you want to know the "what," big data analytics will tell you this; if you want to know the "how" and "why," market research is required. Big Data and analytical methods are used to enhance marketing research. In the hands of those who understand how and when to fully utilize both, "Big Data" and marketing research together offer a powerful toolkit. Market research and big data must always be utilized in conjunction, never as a substitute. Both have distinct goals, advantages, and even difficulties. But when used appropriately, they can be effective resources for analyzing issues with marketing decision-making.

13.6 GAINING INSIGHTS FROM BIG DATA

As we have seen in the previous section, the emergence of big data is gradually reshaping and complementing the research process. Big data implies high dimensionality and a significant sample size. These two characteristics present three special difficulties: High dimensionality causes issues like noise accumulation, erroneous correlations, and incidental homogeneity; high dimensionality combined with a large sample size results in issues like high computational cost and algorithmic instability; and the massive samples in big data are typically combined from various sources at various times using various technologies. We must therefore design more flexible and reliable processes because this raises concerns about heterogeneity, experimental variance, and statistical biases. To process big data, big data analytics combines technologies including database and data mining tools, as well as analytical approaches, to evaluate large-scale, complicated data for a variety of applications designed to improve business performance in a number of ways. The Process for Gaining Insights from Big Data is exhibited in Figure 13.2.

Figure13.2 Process for Gaining Insights from Big Data**Acquisition and Recording**

In the case of big data, information comes in variety of forms and formats. Any type of structured or unstructured source may provide big data. The collection of all the data is the initial phase. In order to generate and enter data for big data systems connected to social media networks, Internet crawling algorithms, user device sensors, financial transactions, etc., data acquisition often starts with hardware and software tools. In hyper scaled computing systems, new data is captured and stored using predefined data acquisition frameworks, such as software tools or protocols.

Extraction and Cleaning

The data that has been compiled may not be meaningful or even ready for processing. It must restructure the data such that it can be processed by machine learning. The process of changing the format of distributed data for additional processing is known as data extraction. The majority of the time, it involves potentially unreliable unstructured data in a variety of formats; for instance, tracking devices like smartphones or wearable fitness trackers may malfunction, social media profiles may be out-of-date, and data files may be damaged by defective storage media. Data cleaning is the process of locating and correcting data issues such as missing, erroneous, or inconsistent values. Data file quality standards are established using error detection and data repair methods, which is important to guarantee that data files adhere to the defined quality standards.

Integration, Aggregation, and Representation

The integration of heterogeneous datasets with different dimensionalities into a single dataset created to meet standardization and analytical requirements is necessary for big data analysis. Identical data, such as financial transactions, may be stored across multiple datasets that are under the ownership of various firms. For instance, a merchant and a credit card

company might both record the same transactions. To conduct accurate analysis, a dataset created from information from the card company and the retailer must be reconciled.

Modeling and Analysis

A suitable machine learning model must be created in order to make predictions about insights and future trends based on the properties of big data. What kind of input you have and what kind of output you need will determine the appropriate model. You can also put a few chosen models into practice to determine which one yields the most accurate results for the available data set. The predictions must then be verified to ensure their accuracy. By assessing the performance of the various machine learning models, you can find the ideal model for the most accurate data insights. At the end of this process, you should have the optimum machine learning model and its precise data predictions. Researchers can create models of people's attitudes and sentiments by applying audio, video, and other analytics to user-generated information found on social media channels. Finding significant patterns in big data requires the application of several different forms of analytics. Human-generated text can be transformed into useful summaries using text analytics, such as customer brand rating tweets. Information is derived from unstructured audio, such as audio-recorded customer complaints, by the use of audio analytics. Analyzing video content similarly allows for the extraction of information from videos like user-created unboxing videos. To investigate the content and architecture of social networks, social media and network analytics are utilized. By using two different types of analytics; community detection, which is a technique for spotting social communities by analyzing user attributes and topic/brand features; and social influence analysis, which can show how people recognize peers and modify their behaviors, it is possible to better understand the interactions and relationships between consumers, businesses, and brands.

Interpretation

Decision-makers must be able to comprehend and analyze the findings after analysis and modelling are finished. Results are translated into decision trees or visual forecasting models to help all parties involved understand the data and improve business decisions. Data visualization may clarify information, bring forth hidden patterns, and forecast behavior.

13.7 APPLICATION OF BIG DATA IN MARKETING

Even though the possibility exists, having big data does not always translate into better marketing. Big data might be viewed as your secret ingredient, raw material, or necessary component. The data themselves are not what's so crucial. Instead, your decisions, actions, and the insights you draw from big data are what truly matter. Marketing firms can have a significant impact in the following major areas by combining big data with an integrated marketing management strategy:

Better Customer Relationship Building: Understanding the customer and their preferences for products and services empowers marketers in decision-making and strategy formulation for customers before they decide on a specific brand. By understanding their tastes and preferences, marketers can plan a smooth customer journey, thereby delivering a better customer experience. If the data on an online booking website state that a particular consumer is a frequent flyer on a particular route, based on this information, the service providers can develop strategies to promote nearby tourist destinations, hotel bookings, etc. Similarly, frequent flyer rewards can also be offered to build loyalty towards such a website. Suggestions may also be provided regarding the lowest fare rates for booking during a specific duration, etc.

Product Development and Brand Positioning: Big data is used by businesses to predict customer needs. They create predictive models to identify customer preferences and offer appropriate products and services. The use of big data simplifies the positioning of products or brands by simply assisting marketers in categorizing various products based on their features. Marketers use a segmentation-targeting-positioning (STP) strategy to effectively position their products in a market place, whether it is a physical or virtual market place. By understanding the tastes and preferences of customers, businesses can effectively position their brands for the right target group. For example, big data is used by businesses like Netflix and Procter & Gamble to predict client demand. By categorizing important characteristics of previous and present products or services and analyzing the relationship between those characteristics and the commercial success of the offerings, they create prediction models for new goods and services. Additionally, P&G plans, produces, and launches new goods using data and analytics from focus groups, social media, test markets, and early store rollouts.

Knowing why a brand is well-liked by a particular consumer segment can help in identifying the niche and in developing a plan that represents the brand and attracts the interest of further consumers. The positioning of a brand can be done based on a variety of factors, including quality, features, status, price, etc. The data collected using big data techniques, when used in the right direction, improves the sales and also encourages healthy competition in a segment.

Designing Promotional Plans: Big Data is made up of information gathered from various social media and e-commerce platforms. Marketers can use this data to analyze and understand the market trends so that a competitive promotional plan can be designed to achieve the marketing objective. For example, Amazon is the top e-commerce retailer in the world. One of the key ways that Amazon uses big data in its marketing to improve sales and customer experience is with dynamic pricing. Dynamic pricing is a tactic you will be familiar with if you have ever bought an airline ticket. When you check prices for a ticket, they go up over time because if you keep checking, this indicates to the airline that you are motivated to purchase the flight and may be willing to pay more. This is a tactic that Amazon applies to its website. They change their prices over 2 million times a day based on data such as shopping patterns, competitor pricing, and how common

the product is. Netflix is possibly the largest online streamer of TV shows and movies, and it owes its success to big data. Netflix uses its big data to provide an unparalleled personal experience that has led to a 93% user retention rate. Netflix collects data such as time spent watching a movie, episode or series, if they binge-watched it or if it took a long time to finish it; where users paused shows; if they resumed after pausing; and so on. All of this data and more is used to create personalized experiences for their users. Recommendations for shows and even a future goal of AI-generated personal trailers are what keep Netflix at the top of the streaming game. (Source: Team Streams, November 15th, 2021)

Price Optimization: Big data gives businesses access to information on the prices of competitive brands. The implementation of a price filter on an e-commerce website enables marketers to comprehend consumers' propensity to spend money on their chosen brands and products. Additionally, information on pricing changes can assist a business in developing future plans to sell off old product inventories or determine the appropriate price for new products to enter the market.

Know Your Customer (KYC): Know your customer-KYC is a popular concept initially conceived in banking and financial institutions. KYC data provides a 360-degree view of customers, which is collected through Big Data techniques. The 360-degree view from big data allows marketers to present customer-specific content when and where it is most effective to improve online and in-store brand recognition and recall. A customer-driven marketing strategy certainly provides a greater return per customer and increases the retention rate.

Improved Customer Acquisition and Retention: This is another benefit of the application of Big Data in marketing. According to a McKinsey survey, companies that use consumer analytics extensively are 23 times more likely to beat their competitors in terms of acquiring new customers. Big Data can assist marketers in gathering and analyzing data from a variety of sources, including the web, mobile applications, email, live chat, and even in-store interactions. It provides the results in time and cost-saving, thereby optimizing marketing performance. Elsevier, the world's largest provider of scientific, technical, and medical information, publishing 430,000 peer-reviewed research articles annually, uses big data to streamline a marketing calendar. Big data and a multi-cloud environment provide an efficient way to closely track journals and books throughout their lifecycle and more effectively schedule resources to streamline production and support marketing. Those articles come from a wide variety of resources across the global organization. Combining big data from multiple clouds and sources across the globe merges many regional marketing efforts into a single global marketing message strategy.

Operational Efficiency: The biggest impact of big data is being felt in operational efficiency, even if it may not always be in the news. Big data analysis and assessment can be used to evaluate production, customer feedback and returns, and other aspects to decrease outages and foresee future demands. Big data can also be utilized to make decisions that are better aligned with the demands of the market.

13.8 CHALLENGES IN APPLICATION OF BIG DATA

The data collected for any purpose will be of no use if it has not been properly analyzed for the purpose of decision making. But there are certain challenges of Big Data which are encountered by the companies.

Lengthy processes: Collection of data in real-times provides the accurate data to the companies. But this process is quite lengthy as there are constant up-dations in data collection. Thus, this may lead to delay in analysis of data.

Designing and Managing Big Data Systems: Users frequently face difficulties while creating a large data architecture due to problems with processing capacity. Big data solutions need to be customized to meet the unique requirements of a firm, which is a do-it-yourself project that calls for IT and data management teams to put together a customized set of technologies and tools. Similar to how relational software engineers and database administrators generally operate, big data solutions likewise demand new abilities for deployment and management.

Lack of proper understanding: Users may not be fully aware of analysis, processing, storage or even the importance of data. Others may not have a clear understanding, despite data analysts' potential expertise. In such a situation, users may not preserve the backup of sensitive data if they do not comprehend the value of data storage. They might not use databases for storage in the right way. As a result, crucial information won't be accessible for quick retrieval.

Storage of data collected: large storage needs are one of the biggest obstacles to using big data. Companies' data centers and databases are storing an ever-growing amount of data. The majority of the material is unstructured and comes from many sources, including text files, movies, audios, and documents. As is well known, unstructured data requires a significant amount of storage space and lacks specialized analytical tools.

Selection of right tools for analysis: As the users do not have accurate idea about the data, they fail to select right tools and techniques to get the desired output. Although there are various tools available for analysis, users frequently choose the wrong technology and make incorrect conclusions. Money, time, effort, and working hours are lost as a result.

Inadequacy of data analytics professionals: The use of big data analytic tools necessitates the employment of highly qualified professionals who are accustomed to working with the tools and deciphering sizable data sets. Lack of such skills puts businesses in a position where they must figure out how to analyze the data and come up with accurate solutions.

Data security: As the era of digitalization is growing leaps and bounds, the security of such a huge data base is another challenge. If proper security applications are not available with the companies, it is easy for data hackers to hack the data and misuse it.

Integration of data from various sources for desired output: Data in a business originates from a variety of sources, including social networking

pages, ERP software, customer logs, financial reports, e-mails, presentations, and employee-created reports. It is difficult to combine all of this data to create reports. For analysis, reporting, and business intelligence, data integration is essential, thus it must be perfect.

13.9 SUMMARY

In this unit, we have discussed the various aspects of big data and its applications in marketing research. We discussed types of big data that include structured data, unstructured data, semi-structured data, and quasi-structured data. Big data is characterized by six Vs: volume, variety, velocity, veracity, value, and variability. Big data implies high dimensionality and a significant sample size. These two characteristics present three special difficulties: High dimensionality causes issues like noise accumulation, erroneous correlations, and incidental homogeneity; high dimensionality combined with a large sample size results in issues like high computational cost and algorithmic instability; and the massive samples in big data are typically combined from various sources at various times using various technologies. We must therefore design more flexible and reliable processes because this raises concerns about heterogeneity, experimental variance, and statistical biases.

13.10 KEY WORDS

Big Data: can be defined as data that has a wider variety, arrives at a faster rate, and arrives in higher quantities

Quasi-structured Data: falls between semi-structured data and unstructured data.

Semi-structured Data: is a combination of both structured and unstructured data.

Structural Data: is defined as data that can be collected, saved, retrieved, and processed in a specific format.

Unstructured Data: lacks any kind of established hierarchy.

Value: means deriving useful information from data using specialized technologies to accomplish a certain purpose.

Variability: is a term used to explain how quickly and how much the data under inquiry is changing.

Variety: refers to the diversity of sources of data, both structured and unstructured.

Velocity: refers to the speed of data generation how fast this data is generated and processed to meet customer demands and requirements in real times

Veracity: is a term used to describe how accurate and reliable a set of data is.

Volume: refers to the size of the data sets that need to be analyzed and processed, which are frequently larger than terabytes and petabytes.

13.11 SELF-ASSESSMENT QUESTIONS

1. Discuss the role of big data in marketing research.
2. What do you mean by “big data”? Explain the types of big data.
3. Describe the characteristics of big data with an example.
4. Discuss the distinct advantages of big data in marketing research.
5. Explain the challenges associated with the application of big data.
6. What are the areas of application of big data in marketing research? Discuss.
7. Discuss the various stages involved in the process of gaining insights from big data.

13.12 PROJECT QUESTION

Social media monitoring involves actively gathering, organizing, and analyzing social media data to learn about brands, products, and services. Choose a well-known brand, you are familiar with and monitor social media sites at regular intervals over the next week to gain consumer insights. Comment on your findings.

13.13 FURTHER READINGS

- 21 Scary Things Big Data Knows About You (forbes.com)
- <https://www.forbes.com/sites/ellenhuett/2015/06/05/how-airbnb-uses-big-data-and-machine-learning-to-guide-hosts-to-the-perfect-price/?sh=17cd00bc6d49>
- How Amazon uses Big Data? | Analytics Steps

UNIT 14 INTERNET BASED MARKETING RESEARCH

Learning Outcomes

After going through this unit, you should be able to:

- appreciate the purpose of online marketing research.
- explain the applications of internet-based marketing research.
- discuss the advantages and limitations of online marketing research.
- describe the steps involved in conducting online marketing research.
- explain the various techniques used for online marketing research.

Structure

- 14.1 Introduction
- 14.2 Purpose of Online Marketing Research
- 14.3 Applications of Internet-based Marketing Research
- 14.4 Advantages and Limitations of Online Marketing Research
- 14.5 Steps in Online Marketing Research
- 14.6 Online Marketing Research Techniques
- 14.7 Summary
- 14.8 Key Words
- 14.9 Self-Assessment Questions
- 14.10 Project Question
- 14.11 Further Readings

14.1 INTRODUCTION

The development of the Internet has given businesses of all sizes and types access to the information they need to remain competitive. The Internet significantly improves a searcher's capacity to find difficult information in local or global markets. Many of the standard print resources and the online database systems that required separate dial-up access are now available online. However, it should be noted that while the Internet can rapidly address some urgent information needs, it also tends to result in information overload. The researchers must present their own perspectives on the Internet's usefulness and application to the tasks at hand, given the time-limited resources available. In this unit we will discuss the purpose, applications, advantages, techniques, and trends of internet-based marketing research, including how to use existing sources, online surveys, chat-based sessions, online focus groups, social media listening, and more; so that both large and small companies can take proven traditional market research techniques and adapt them to the Web.

14.2 PURPOSE OF ONLINE MARKETING RESEARCH

Internet has become increasingly used platform for carrying out the market research. It is a major source for market researchers due to the availability of massive information. In recent years, there has been a substantial growth in internet users worldwide. Accessing the information through internet is easy and efficient as compared to the reliance on traditional modes of data collection. In countries like India, where internet is readily available, consumers participate in online polls and surveys and share their experience towards product or a service. This enables marketers to easily update their consumer databases and use this information while they conduct internet research. A method of gathering data based on the usage of the internet is known as internet-based market research. Internet-based market research is also known as online marketing research. It involves creating online surveys, collecting data, analyzing it, and maintaining an online database for future use.

The technique through which firms collect data online to determine how well a good or service is accepted by consumers is known as internet-based marketing research. By analyzing this data, further helpful information can be obtained, such as current trends, which can aid a business in creating a more effective strategy.

Conducting marketing research on internet could be a complex process. It may also require a considerable level of expertise in obtaining data. Sometimes, the customers shopping online may feel reluctant to share the information or they may get biased while recording their opinions. Another issue behind this element is confidentiality that the user's personal information may be violated. As internet is widely available and customer demographics can be easily segmented, marketing research on internet is beneficial due to many reasons. According to Pitts et al. (1996) Web sites have been utilized to accomplish the goals of a number of marketing strategies, such as:

1. To get a deeper understanding of company's target audience
2. To understand online consumers' tastes, preferences and the buying habits
3. To discover new business opportunities and business models
4. To understand the recent trends on e-commerce platforms
5. To identify potential areas of growth
6. To determine existing customer satisfaction level and gather feedback.

14.3 APPLICATIONS OF INTERNET-BASED MARKETING RESEARCH

The Internet can be used by marketing researchers for a variety of purposes, such as an alternative tool for data collection or to analyze online consumer behavior. Through or with the help of the Internet, a wide range of marketing research tasks can be carried out. There is a plethora of information on the

internet, some of it accurate and some of it not. Since there is minimal control over the material provided, it is important to verify the validity and accuracy of the data gathered online. To guarantee that the data has been legally collected and is fully referenced, the source from which it was obtained should be verified. Checking the date that the material was published online, or when it was submitted to a website, is crucial because not all websites update frequently. In addition, companies should make sure that their studies won't harm their reputation or trigger legal action.

Let us discuss some of the major applications of Internet-based marketing research.

Analysis of the marketing environment

With the help of the internet, businesses may readily keep track of their external environment. Search engines like Google, Yahoo, and Bing are a good source of information for businesses. Companies are kept up to date on news developments as they happen using RSS (really simple syndication) feeds from news websites. The opinions of various individuals on a number of topics can be found in online forums and blogs. Research can be found on the internet from marketing research organizations which conduct market research about consumer trends covering a diverse range of sectors and industries.

Analysis of Competitors

Companies can use the internet to research competitors and evaluate their marketing mix strategies. Many companies use websites to draw customers to their businesses. These websites offer a lot of useful information. A company's website often includes a thorough catalogue of its items, together with frequently available pricing information. Business websites also include details on a company's product and service promotion strategies. You can also acquire financial reports outlining how a business is operating through the corporate website or another website that charges customers to access company financial reports.

Online Surveys

Online surveys are an easy way to do primary research through the internet. A notice requesting participation in an online survey will appear when an internet user accesses a website with online questionnaires. Users will be questioned about their opinions of the business, its products and services, its websites, and the industry in which it competes. Online surveys are often brief and easy to complete to encourage user involvement. Analyzing the data received from online surveys is used to develop a marketing strategy.

Analysis of Usage

Businesses do research to learn about customer buying habits and how popular their websites are. Some organizations use their own software to acquire data on internet usage, while others hire outside firms to carry out the investigation. Usage analysis will reveal the number of visitors to the website, where they came from, how long they stayed there, and which sections of the website they used.

Cookies.

Small text files known as cookies are momentarily downloaded to a user's browser each time, they visit a website. Cookies provide website owners with information about your visits and usage patterns. Cookies keep track of which websites are visited, how long visitors stay on each page, and how they move about the website. Businesses can enhance their websites and marketing by using the data gathered through cookies. Additionally, cookies enable websites to be customized for users who have registered with them.

As A Source of Secondary Data

The Internet plays a bigger part in marketing research than just gathering primary data. If not exclusively, the Internet serves as a repository for secondary data. There are numerous benefits to using secondary data from the Internet. Certainly, one of them is the simplicity of data access. Data about particular topics is accessible around-the-clock from any location on Earth. While many institutions offer test results for free, others frequently post the summary or findings on their websites, even if they charge for it. However, the issue with using secondary data is the potential for a mismatch between the needs and the available data. This is because information may be gathered for purposes that are not explicitly stated or even unrelated to the interests of the subject.

Activity 1

Discuss some examples of how the Internet may be used as a secondary source of data for marketing research.

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14.4 ADVANTAGES AND LIMITATIONS OF ONLINE MARKETING RESEARCH

1. **Wider geographical reach:** Use of internet for marketing research allows the researchers to wider the scope and collect the diverse feedback across the geographical borders.
2. **Faster data collection:** Allows the researcher to collect more data in lesser time compared with the traditional method of data collection.
3. **Varied segments across geographical locations:** Researchers while using internet-based techniques for the purpose of data collection may go deep into segmentation across the geographical locations.
4. **Immediate outcome:** Marketers while conducting the online marketing research and collect and acquire the data instantly for the purpose of analysis and review as collection of feedback online involves lesser time as compared to traditional methods.
5. **Customization:** Unlike traditional methods, it is possible for the researchers to customize the market research survey depending upon

the demographics, language, cultural values and other preferences of the online participants.

6. **Cost effective:** Market research data may now be obtained more quickly and affordably thanks to the internet, email, and the rise of mobile devices. Online surveys are far less expensive than phone calls, in-person interviews, or surveys sent by postal mail for conducting market research.
7. **Diversity of Questions:** Market research online surveys can be designed with a variety of question types and logic, including multiple-choice questions, single or multiple responses, open-ended questions, multimedia questions, rating scales, dichotomous yes/no questions, and built-in advanced logic conditions, thanks to the abundance of research support and software that is available online.
8. **Easy to Use:** Online surveys for market research save time for researchers. Data analysis is made simple by the accessibility of the collected information, which can also be quickly transformed into spreadsheets, graphs, and reports.

The advantages and limitations of conducting Internet-based marketing research is given in table 14.1

Table 14.1: Advantages and Limitations of Internet-based marketing research

Advantages	Limitations
– Relatively low cost – Easy application and speedy execution - Wider geographical coverage – reaching out to a large group of respondents at the same time – the capability to automatically generate transcripts for qualitative research and to process quantitative data in real-time – customization of instrument depending upon respondents' previous opinion e.g., interviews on chatbots	– Difficulty in application in the areas with low internet penetration - Not suitable in case of weakness of ICT infrastructure, hardware barriers to potential respondents, – Anonymity of expression or dummy responses in place of the original participant may lead to biasness - Due to the lack of direct interaction, there is little control over whether the chosen respondents actually participate in the survey. – Lack of participant evaluation of the sensory properties of the products. – lack of opportunity to observe the verbal and non-verbal expressions and reactions of the participants

Source: <http://www.worldscientificnews.com/wp-content/uploads/2016/06/WSN-57-2016-712-721.pdf>

14.5 STEPS IN ONLINE MARKETING RESEARCH

Define the target audience

The target group defines the customer segment whom you are targeting for the entire campaign. The marketers need to define the online customer demographics such as age, geographical location, income and spending pattern, habits etc.

Preparation of research question

Once the target group has been identified, it is required to design a relevant technique and prepare questionnaire for the purpose of data collection. E.g., if the marketer wants to find out consumers' preference to a specific service on mobile application, say, Urban Clap, then the respondents involved in data collection must be the actual users of this application.

Ensure the active engagement of the participants

Making a plan for how to connect with your target audiences comes next. This could come through a number of marketing methods, such as social media campaigns, email campaigns, direct customer outreach, or online research portals that can assist in disseminating the survey to the appropriate target audience. The aim is to ensure that you connect with as many relevant audiences as you can.

Summarize the findings and report preparation

Once the results are achieved, the researcher needs to summarize the findings based on the recorded opinion of the participants for the purpose of analysis and report preparation. The conclusions must to include an executive summary, findings and an action plan, participant background information, and thorough explanations of what the findings mean.

14.6 ONLINE MARKETING RESEARCH TECHNIQUES

Online Surveys: This is the most commonly and popularly used tool to collect data. A set of questionnaires is sent out to the target customers via multiple channels, including email, websites, social media, etc. Online surveys are used by marketers to get information and opinions about their products and services. Similarly, the effectiveness of various marketing strategies can also be measured with the help of customer feedbacks through survey.

An efficient survey must include different types of open and close ended carefully designed questions. An online survey should include questions that are appropriate in collecting the needed information from the chosen sample. Only when the survey's goal is supported by the questions used, will a researcher be able to effectively collect data through online surveys. For

instance, marketers may use dichotomous questions to record customers' satisfaction in the form of agreement or disagreement or Likert Scale-type questions to determine the respondents' level of agreement with a specific statement. Researcher must understand the purpose of survey while designing the questions while using it for data collection. For example, Online food delivery apps like Zomato asking for service rating.

Online Focus Groups: Similar to the Focus Groups technique used in traditional data collection method, online focus groups are good for gathering qualitative data directly from the participants. Online focus groups can be divided into two parts.

Chat based focus groups: Also called as Asynchronous focus group. Users respond to the surveys, questionnaires or even post the pictures, videos in the form of their feedback to the questions. Participants can also interact with each other with the help of a moderator. For example, an apparel brand that launches a new clothing line and then takes feedback on various aspects such as pricing, store ambiance, product trends, etc.

Real time focus groups: This focus group discussion takes place in real-time at a predefined time that is already communicated to the participants. It usually consists of 8-10 participants who discuss on predefined objective with the help of moderator. The qualitative data collection is done during the discussion.

The advantages of online focus groups are they conducted anytime and anywhere. Online focus group monitoring is also easy and the moderator can continuously monitor qualitative insights from members. Secondly, there are no physical, geographical or demographic restrictions with online focus group researches. Thirdly, the data collected through this technique can also be analyzed using online tools available. Various marketing areas where online focus group research is used are advertising and brand research, new product development, service evaluation, consumer research etc.

Online Communities: Also called as internet communities is a platform where participants interact through various social media platforms, chat rooms, mailing lists, discussion boards and other forums. An online community is simply a group of people coming together for a common purpose, interest, or vision, and doing so via the internet. The 'family members' of the community share the ideas with common interests. Such communities are popularly found in gaming applications. These family members can post, comment, discuss, give advice or collaborate as well. e.g., Facebook, Twitter, Instagram etc. People may also join online communities through video games, blogs, and virtual worlds, and could potentially meet new significant others in dating sites or dating virtual worlds.

These communities encourage the continuous development of peer-to-peer interaction. Participants can join without any geographical limitations. The closeness this platform provides enables accurate and actionable insights to be drawn from natural conversations between participants in online spaces such as forums and blogs. These platforms can be accessed via mobile or desktop and are monitored by researchers for both safeguarding and insight generation purposes, so the space remains safe and enjoyable for all involved.

This technique can be used as a standalone method, or in conjunction with scheduled research tasks to enhance the value of the insights generated by allowing further informal discussions outside of the task.

Google Analytics and Google Keyword Tool: Keywords are words or phrases that are used to match the ads that the people are searching for. Selection of right keywords helps the marketers to reach to the customers during the ad campaign. The Google Keywords tool acts as a window into the behavior of consumers when searching online for products or services. Google Analytics is a free tool provided by Google to understand the website traffic. Google Analytics provides feedback as to how the customers are behaving on company's website. It may show the marketers those products many views, but few buy (or vice versa) and it can illustrate what social media channels customers are responding to, amongst many other insights. Google Analytics puts several lines of tracking code into the code of user's website. The code records various activities of audiences when they visit website, along with the attributes (such as age, gender, interests) of those users. It then sends all that information to the GA (Google Analytics) server once the user exits website. Google Analytics aggregates the data collected from marketers' website in multiple ways, primarily by four levels:

1. User level (related to actions by each user)
2. Session level (each individual visit)
3. Pageview level (each individual page visited)
4. Event level (button clicks, video views, etc.)

Social Media Listening: Social media listening takes the sentiment underlying the data into account. Social media sentiment analysis is a crucial component of social media listening in order to understand how people feel about you and your competitors. In order to achieve significant commercial results, you should focus on what you can learn from social dialogues rather than simply tracking the number of times your brand is referenced. You can react swiftly to unforeseen events by being aware of shifts in societal sentiment.

Online Interviews: Face-to-face conversations are possible during online interviews, so you may let the conversation flow naturally and keep an eye on the interviewee's body language. Your interviewees can respond to self-reflection questions to aid in the creation of your buyer profiles. These buyer profiles outline the age, number of children in the family, income level, profession, difficulties encountered at work, and other characteristics of your ideal customer's lifestyle. Your whole marketing plan, from the features you include in your product to the information you publish on your website, can be influenced by having this buyer profile in hand.

Customer Satisfaction and Loyalty Surveys: Customer satisfaction and loyalty surveys will help you understand how to get existing customers to do business with you again and what will inspire them to do so (such as loyalty programs, and rewards.). You can use the results of this research to figure out how to best encourage customer satisfaction.

Blogs: A blog is a subsection of a company website that is regularly updated by adding new contents. It enables to interact with readers more, either by tracking how frequently readers share your blog posts on social media or by enabling reader comments on specific blog posts. In this regard, a blog differs from the rest of your website in that it is more like a two-way communication. However, a blog can also be a whole website and is commonly the case if it is primarily meant for personal use, like a travel or fashion blog.

14.7 SUMMARY

In this unit, we have discussed the purpose, applications, advantages, techniques, and trends of internet-based marketing research, including how to use existing sources, online surveys, chat-based sessions, online focus groups, social media listening, and so on. The Internet significantly improves a researcher's capacity to find difficult information in local or global markets. Many of the standard print resources and the online database systems that required separate dial-up access are now available online. However, it should be noted that while the Internet can rapidly address some urgent information needs, it also tends to result in information overload. Additionally, anonymity of expression or dummy responses in place of the original participant may lead to biased results.

14.8 KEY WORDS

Blogs: A blog is a subsection of a company website that is regularly updated by adding new content.

Cookies: Small text files known as cookies are momentarily downloaded to a user's browser each time, they visit a website. Cookies provide website owners with information about your visits and usage patterns.

Online Communities: also called as internet communities is a platform where participants interact through various social media platforms, chat rooms, mailing lists, discussion boards and other forums.

Online Focus Groups: are used for gathering qualitative data directly from the participants.

Online surveys: are an easy way to do primary research through the internet.

14.9 SELF-ASSESSMENT QUESTIONS

1. Discuss the purpose of online marketing research.
2. Outline the applications of Internet-based marketing research.
3. What are the limitations and benefits of online data collection? Explain.
4. What are the steps in collecting data through internet-based techniques? Discuss.
5. What are various techniques of online data collection?

14.10 PROJECT QUESTION

Conduct an online survey to measure the smart phone usage behavior of your friends.

14.11 FURTHER READINGS

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UNIT 15 MARKETING RESEARCH AND SOCIAL MEDIA

Objectives

After going through this unit, you should be able to:

- appreciate the role of social media in marketing research.
- explain the benefits of using social media for marketing research.
- outline social media applications for marketing research.
- describe the process of social media marketing research.

Structure

15.1 Introduction

15.2 Role of Social Media in Marketing Research

15.3 Benefits of Using Social Media for Marketing Research

15.4 Social Media Applications for Marketing Research

15.5 Social Media Marketing Research Process

15.6 Social Media Research Insight

15.7 Summary

15.8 Key Words

15.9 Self-Assessment Questions

15.10 Project Questions

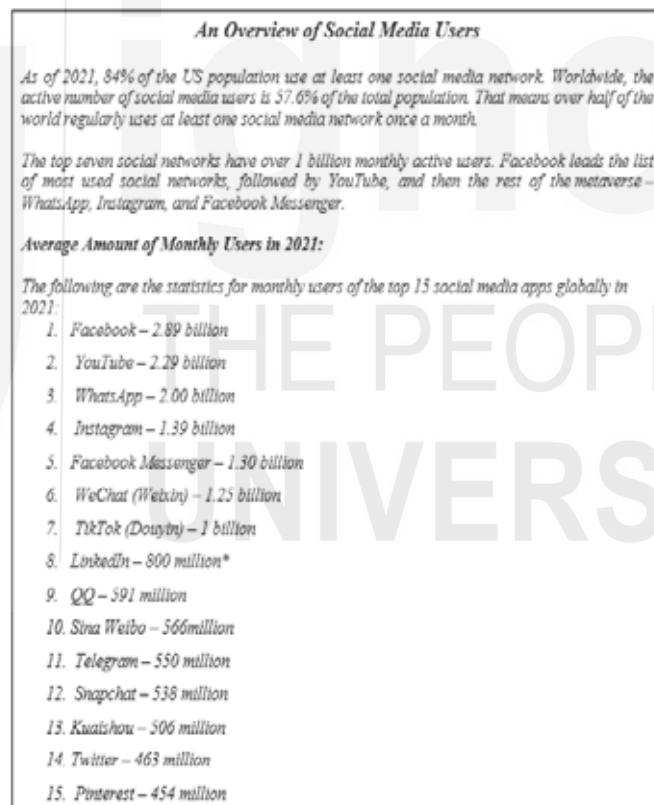
15.11 Further Readings

15.1 INTRODUCTION

Promoting a product or service through various social media channels has become an intense need of today's organizations. Many organizations have realized that it is a must to go for social media if they want to build a strong brand and remain competitive at a market place. The question is to what extent social media is suitable to reach and develop a connect with target audience? Here, social media in marketing research plays a vital role in developing an understanding of research objectives. Due to constant elevation of technology, organizations are constantly leveraging various marketing research techniques by using various social media platforms, which are efficient and cost effective as well. Social media provides and insight on their brand's goodwill, customer tastes and preferences, target market characteristics and other important facets required for market research. In this unit, we will discuss the various aspects of social media from a marketing research point of view.

15.2 ROLE OF SOCIAL MEDIA IN MARKETING RESEARCH

With the widespread use of mobile devices and apps, consumers have round-the clock access to the information superhighway. These digital advances have given consumers the ability to produce their own content on social media. Social media is characterized by user-generated content. Different social media platforms have specific features that make them useful in marketing research. A key element in the success of social media is the ability of users to evaluate, rank, comment on, review, and reply to the new media environment. People combine one-to-one (like email and instant messaging), one-to-many (like Web pages and blogs), and many-to-many (like wikis) communication modalities to create online communities. Social networking sites like Facebook, video sharing sites like YouTube, photo sharing sites like Flickr, wikis like Wikipedia, product and service reviews like TripAdvisor, virtual worlds like Second Life, Web-based communities, blogs and microblogs like Twitter are some examples of social media.



Source: <https://www.searchenginejournal.com/top-social-media-statistics/418826/#close>

These social networks and the open-source social computing tools on which they are based can be used by marketing researchers to broaden the scope of their research offerings and allow companies the opportunity to interact with their customers. As a result, they are applicable to a range of marketing research applications, including as segmentation, idea generation, concept testing, product development, brand launches, pricing, and integrated marketing communications.

In order to “listen” to and interact with customers on social media,

companies can benefit from marketing research. Social media platforms provide fresh perspectives on how to comprehend, justify, influence, and forecast customer behavior. For example, Cadbury Chocolates chose to discontinue production of the Wispa candy bar for UK market in 2003. However, the Wispa had a sizable following that could “speak” through Facebook. Around 93 Facebook groups with over 14,000 members each demanded that Cadbury bring back the Wispa. In 2007, Cadbury responded to these customers’ requests and brought the candy bar back. In the case of Cadbury, active listening and teamwork were successful. 41 million bars were sold out in just four weeks after the product’s debut, and millions more have since been sold.

To know the meaning and importance of social media, it would be better to compare social media to traditional media, as exhibited in Table 15.1.

Table 15.1 The Differences between Traditional and Social Media

Traditional Media	Social Media
Fixed, unchangeable	Instantly updatable
Commentary limited and not real time	Unlimited real-time commentary
Limited, time-delayed bestseller list	Instant popularity gauge
Archives poorly accessible	Archives accessible
Limited media mix	All media can be mixed
Committee publishers	Individual publishers
Finite	Infinite
Sharing not encouraged	Sharing and participation encouraged
Control	Freedom

Saylor URL: <http://www.saylor.org/books>

The role of social media in marketing research is to collect quantitative and/or qualitative data from various social media platforms in order to analyze distinct trends. It is a responsibility of marketers to understand and know their target audiences’ needs and preferences. Social media market research simply means use of social media platforms for gathering all time data to develop a better understanding of company’s target market. As millions of conversations take place on social media, marketing research organizations have developed the capacity to “listen” to them and give clients information about their brands and businesses. These applications are known by a number of terms, including sentiment analysis, text analytics, brand monitoring, and buzz mining. Examples of marketing research companies that offer these services are Decooda, Conversition, BuzzMetrics, and so on.

#SpeakBeautiful

A survey found that women made five million negative tweets about body image and beauty in 2014 alone, making them twice as likely to criticize themselves on Twitter as they are to compliment themselves. In a social media-connected society, these unfavorable discussions—from friends, family, celebrities, and advertisers—fill feeds and minds with self-doubt, impeding women's paths to beauty confidence.

Dove chose to launch a campaign based on the idea that these kinds of tweets caused self-doubt in people's minds and newsfeeds. Dove and Twitter launched the #SpeakBeautiful campaign in 2015 with the goal of changing this reality, influencing behavior change, and igniting a cultural revolution in the way women talk about beauty online. The Oscars were chosen as the unveiling, due to much discussion around beauty. During the red-carpet ceremony, Dove unveiled a 30-second commercial to begin the campaign.

Dove was the first company to use Twitter's Reverb product as part of their relationship, which gave them the ability to track how frequently #SpeakBeautiful was used on Twitter during Oscar night. Additionally, they developed real-time data visualizations of social data that displayed the most typical negative and positive words women used to describe their appearance on Oscar night. To reach those who were already talking about the Oscars and beauty, Dove used the social influencer arm of Twitter called Niche and personalized ads. Women all around Twitter responded positively to the initiative, offering their own inspiration and making powerful videos showcasing their setbacks and victories. Beyond the Oscars, the #SpeakBeautiful Twitter campaign continued to trend during the MTV Video Music Awards and was incorporated into regular tweets.

Based on the findings of Twitter research that showed women's confidence in their appearance is lowest in the mornings, on Mondays, and on weekends, Dove developed content expressly for these times to help women.

Through targeted Twitter promotion, Dove was able to connect with women throughout these crucial times, bringing about change day by day and week by week. Dove community managers identified real remarks from women regarding beauty and body image using Lithium, a proprietary Twitter technology, and then offered one-on-one support to help them #SpeakBeautiful.

There were indications that the campaign succeeded in achieving its objectives. Negative body image tweets fell by 36.8% from 5.3 million in 2014 to 3.4 million in 2015. When comparing tweets sent during the Oscars in these two years, women posted 69% more self-affirming tweets and 30% fewer negative tweets. Women were motivated by the message. The hashtag #SpeakBeautiful was used more than 168,000 times and generated 800 million social media impressions throughout the course of the campaigns. The #SpeakBeautiful campaign, according to Twitter, increased brand sentiment for the Dove brand by 17%.

Source: Dove and Twitter #SpeakBeautiful - The Shorty Awards

Activity 1

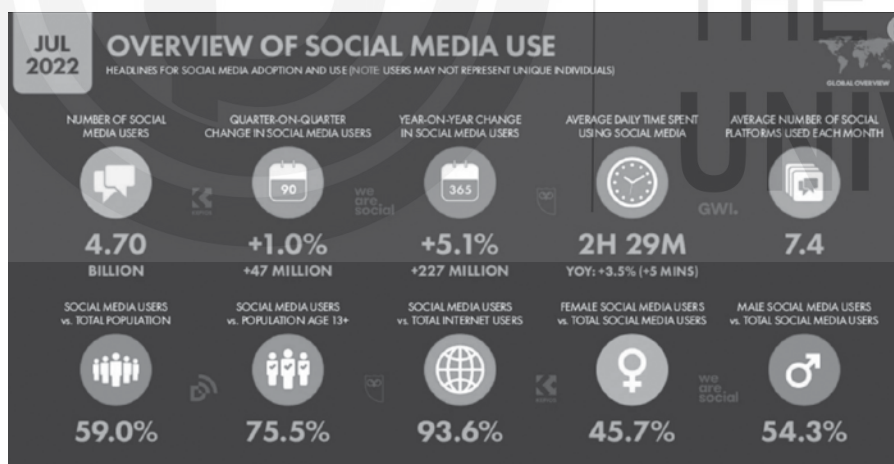
Discuss the role of social media in marketing research for small businesses.

15.3 BENEFITS OF USING SOCIAL MEDIA FOR MARKETING RESEARCH

Social media can supplement or even replace the traditional mode of conducting market research. It plays a vital role in understanding the brand's target audience. According to a survey, during the fourth quarter of 2021 number of daily active users on Facebook were 1.93 billion, with a gain of 84 million users. United States has been recorded as a country with the most twitter users in 2022 followed by Japan with 76.9 million and 58.95 million users. India stands at the third position with 23.6 million users according to Statista Report 2022.

Over the past ten years, social media users have grown at a higher rate than internet users, with a double-digit annual growth rate of 10.1%. In 2022, there will be 4.62 billion social media users worldwide, or 58.4 percent of the world's population (Figure 15.1).

Figure 15.1 Overview of Social Media Use



Source: <https://www.smartinsights.com/social-media-marketing/social-media-strategy/new-global-social-media-research/>

Market research through social media is advantageous for following reasons:

- **Affordability:** Social media channels are the cheaper source for marketing research as compared to other techniques such as focus groups of survey etc. which may require to spend a lot, depending upon the nature and complexity of research objectives.
- **To arrive at faster results in real time:** Traditional market research

methods could be time consuming and may provide the data that has lesser relevancy to the research objective. Social media provides real time and updated output which may help the marketer to form strategy for immediate results.

- To gain and insight towards current data: Marketers get the insights about their products and services in more recent terms. In the sense the changes in consumer preference and tastes over the time can be monitored constantly.
- **To assure more accuracy and promptness:** As the information gathered is shared by the customers themselves, it is unprompted and it more likely to reveal true perception.
- **To broaden the scope of market research:** We all use one or more social media platforms for various reasons. Thus, using social media platforms while conducting market research, marketers can gain an access to large number of audiences as compared to the traditional techniques available for data collection.
- **To understand and define target audiences:** Social media enables the marketers to discover their target audiences in a large fragmented market. Marketers through online discussion platforms can find out the segments with similar interests where they can provide the product or service-related information. This is a common platform to disseminate a piece of information for a larger community at once.
- **To analyze competitors:** Social media monitoring can enable many organizations to understand the competitors' campaigns, strengths and weaknesses. Marketers can find out how the customers are reaching to the competitors' products/services as well as their perception towards various products.
- **To monitor large audience:** A report by Zacrep (2015) on 'How Researchers can Effectively Utilize Social Media Analytics', social media can be used effectively to monitor the target audience. According to Nielson report (2011) report, around 80% or more people who have the access to internet use various social media platforms. This access to masses makes it possible to gather larger data about customers and target audience.
- **To discover the recent marketing trends:** Social media allows the marketers to learn more through observations. Research tools such as online surveys, questionnaires, online discussion forums, blogs etc. can be effectively designed to understand the trends that the customers are following at a market place. Marketers can analyze the collected data to understand customer engagement and even participate in various online discussion forums.



Source: <https://sproutsocial.com/insights/social-media-market-research/>

Activity 2

Describe with an example how social media data helps marketers better understand their target audience.

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15.4 SOCIAL MEDIA APPLICATIONS FOR MARKETING RESEARCH

- **Analytics for Qualitative Content:**

Marketers can use analytics to assess the customer engagement with brands through social media. The likes, comments and shares on social media page helps the marketers to understand the kind of content, product or service that is favored by the target audience. Similarly, marketers can analyze the competitors' performance while comparing it with their own social media page and draft similar strategies to increase brand awareness and customer engagement.

Twitter Analytics-You may get data on likes, responses, retweets,

and other metrics from Twitter Analytics to help you understand the reach and impact of your Twitter ads as well as do research on the effectiveness of your own or a brand's Twitter channels. You can also examine the best and worst posts from any given month. Additionally, Twitter lets you conduct in-depth keyword searches across specified time frames and places, which enables you to gauge public opinion on a subject.

Facebook Analytics- On its analytics page, Facebook Analytics also provides comprehensive, downloadable statistics on your owned Pages. On Facebook Pages, you can view information about your page's demographics as well as reach and engagement statistics for posts and videos. Researching how people interact with public Facebook Pages is another option, however due to Facebook's privacy policies, it is challenging to compile this data.

You Tube- For creators, the analytics section of YouTube offers a range of metrics to look at, including watch times for your videos, traffic sources to determine where viewers are coming from, and demographics reports. On YouTube, you can also conduct additional social media research by looking at the Trending page and examining the views and comments that videos have received.

- **Social Listening:**

Social listening or social media listening refers to the process of identifying and monitoring the conversations or feedback about product, individual or organization from various social media channels. It allows the marketers to understand the voice of customers, their preferences and expectations from a product, service and even an organization. Such data is useful to for improving operational and marketing decisions.

- **Online Opinion Polls and Questionnaires**

To get feedback on consumer preferences and opinions toward an organization's products and services, marketers might use social media to conduct questionnaires or opinion polls. Various social media platforms such as Instagram, allows its users conduct online polls through Instagram Stories. Followers view and comment on these stories. The poll's results and the user's selections for each option are both visible.

- **Opinion Polls:**

It has been observed that the posts that ask questions receive more likes than other posts on social media, especially in case of Facebook. Marketers can create an opinion poll to get valuable feedback and data for further analysis.

- **Contests:**

Organizing contests is one of the popularly used promotional tool by marketers. A change to win a prize motivates the audience to share their details and user generated contents or their valuable feedback on

a post.

- **Call to Action Posts:**

Call to action is a part of a content, ad or a website that prompts the audience to move ahead with the content and take action. It could be a link, pop-up, clickable image or a button. Click to action usually leads to creating an account, write a quote, registering for the course etc.

- **Content Analysis:**

Content analysis is a tool which is used to analyze the data collected in qualitative form. This tool finds where in a text a specific word or phrase appears. To quantify and assess the presence, significance, and relationships of such words, themes, or concepts, researchers employ content analysis. The methods for gathering qualitative data utilized in content analysis include interviews, open-ended inquiries, field study notes, conversations, and any instances of communicative language (such as newspaper headlines, discussions, speeches, and so on). The entire text is categorized into several categories in order to use content analysis to analyze the keywords. The codes can then be further divided into «code categories» to further categorize the data after the text has been coded into code categories.

- ❖ **Advantages of Content Analysis**

- Analyses text-based communication
- Can be applied for qualitative as well as quantitative analysis
- Decoded data can be analyzed using a suitable statistical tool
- Provides an insight into complex terminologies aroused during data recording
- Relatively accurate method if well applied.
- Easily understood and less expensive in application.
- Generates more powerful data if combined with other qualitative tools such as interviews and observation.
- Can be applied to analyze the historical data as well.

- ❖ **Disadvantages of Content Analysis**

- Manually coding the qualitative data could be time consuming.
- May generate results with errors particularly when high level of interpretations is expected out of analysis.
- May not generate effective results when applied on complex texts
- Results sometimes may produce simple word counts
- Sometimes difficult to computerize or automate the data
- Involves certain level of subjective interpretation which may question the reliability and validity of results

Where do we apply Content Analysis Technique?

- To determine a person, group, or organization's intentions, purpose, or communication patterns.
- To analyze the responses based on attitude and behavior
- To identify the psychological or emotional state of persons or groups
- To reveal differences and patterns in communication content
- To examine the information gathered through focus groups, panel discussions, interviews, and open-ended questionnaires in order to supplement quantitative data

Types of Content Analysis:

1. **Conceptual analysis:** Conceptual analysis reveals the presence and occurrence of concepts in a text. In a typical analysis process, a concept is chosen for examination and quantified by counting its presence in a given text. The main goal of Conceptual Content Analysis is to examine the occurrence of a selected term in data.
2. **Relational analysis:** Similar to conceptual analysis, relational analysis starts with the selection of a concept to be studied. But the analysis entails looking at how several concepts relate to one another. It is believed that interactions between concepts, rather than individual concepts, are what give things their meaning.
3. Different outcomes, conclusions, interpretations, and meanings may result from various types of analysis.

For marketing researchers to benefit from the increasing comfort levels seen in social media interactions, Leiman (2010) offered the following four specific recommendations:

- Establish a social networking site for affinity groups so that new users of digital journals can instantly feel at ease contributing information.
- Create a platform for rich digital journaling using technology, including images, video, and other media, and promote regular and impulsive use of it.
- Don't be afraid to request that customers tweet or blog about their habits or personal reflections.
- However, be mindful that, without specific agreements, customers who take part in a marketing research social digital journal study might discuss their experiences on other social media platforms, so letting rivals know about the client's activities.

15.5 SOCIAL MEDIA MARKETING RESEARCH PROCESS

Figure 15.3 Stages of Social Media Marketing Research Process



Step 1: Define the target goals:

Whether to conduct market research on product, service, brand, organization of any aspect related to consumer behavior and insights, both online and in physical form

Step 2: Define method of data collection:

What type of data collection is required: Qualitative, Quantitative or both? Quantitative method includes data collected through structured questionnaires using various scales. In case of online data collection in quantitative form, it involves counting responses in the form of comments, shares, likes, views, number of followers or subscriptions etc. Qualitative technique involves understanding audiences' or respondents' opinions in the form of sentiments, interpretations, observations on a certain issue, word contextualization, picture or theme association, industry trend analysis etc. But a popular text categorization technology known as sentiment analysis can spot the subjective information in a source of information. While observing online chats, it assists researchers in comprehending the social sentiments around their brands, products, and services. It analyzes the message and tells whether the underlying sentiments are positive, negative or neutral.

Step 3: Select a Social Media Channel:

Selection of social media channel is very vital. This involves selecting a social channel depending on the varied demographics across the regions and behavioral norms. Selection of social media channel depends upon research objectives and strategy.

Step 4: Choose the Right Tool and Keywords:

It is important for the marketers to understand a right set of keywords that will help them to achieve the accurate data and shape the results as in detail

as possible. From this data the marketers can track additional information required in relation to a product, understand the quantitative data and formulate marketing strategy to capitalize positive feedback. Marketers may quickly obtain qualitative information from word clouds, which show the most frequent terms used in conjunction with the major keywords.

Step 5: Data Cleaning:

Data cleaning is a process of identifying incomplete, irrelevant or misleading information or part of information that contains in the data set and taking corrective action against it to reflect more accurate and relevant result. This is one of the time-consuming aspects especially in social media research. Many popular social media platforms spend a lot of time data clearing as their names have become the substitute to social media platforms. The data can also be cleaned by excluding large and unnecessary words, applying segments like male, female etc.

Step 6: Draw conclusion and convert into strategy:

The end goal is of course how businesses turn social media market research into strategy. Dashboards offer digital interfaces that let people view information quickly and readily that is presented in a simplified way. Based on information, strategies can be formulated to accomplish the desired goals.

15.6 SOCIAL MEDIA RESEARCH INSIGHT

The usage of social media by students for academic purposes:

Every aspect of life has been impacted by social media and its content, whether it takes the shape of text, photographs, videos, or any other format, thanks to the advancement of media and the internet in our daily lives. The educational sector has a similar situation. In addition to books and teachers who are physically present, technology has created a number of new learning opportunities. Prior to the widespread adoption of technology, children's lives were governed by chalk and duster, actual classrooms, students, and teachers. Traditional schooling was once seen as the best option. One could say that a teacher's value in a student's life cannot be overstated, and this is true in all respects. However, though a teacher's role in a student's life can never be replaced, technology has changed how students interact with a teacher and makes a big difference. Social media and its content, as well as the availability of resources on the internet, have done just that. Social media and the internet have eventually evolved as a new industry of their own, changing the interface of traditional study materials and their distribution.

Today's students can utilize the internet to share knowledge and pick up skills from anywhere in the world without having to leave their home country or even the countryside. One can connect and deliver lectures using the same technology. The way people used to feel about sharing their knowledge has also altered as a result of the advent of blogging and YouTube. To share their valuable information with students around the world, many professors and specialists are turning to these two for assistance.

Divya Sharma and Rajeev Shukla investigated what types of study materials

students choose among academic blogs and YouTube study material in order to assess how students perceive social media for collaborative learning in higher education institutions. They collected samples of 100 undergraduate and graduate students from a university and used an online survey to gather data on some demographic factors, like gender and age, several behavioral factors, including the amount of time spent on mobile devices, how often internet-connected devices are used, how often students search for online study materials, and which social media platforms they prefer for finding online study materials. A number of other variables were also included in the study to measure how students used social media to access online course materials (such as relevancy, easy to use, authenticity of shared information, create learning interest, easier to learn, etc.). Using a five-point Likert scale with the anchor (1=strongly disagree to 5=strongly agree), the statements based on these criteria were scored.

Analysis of the collected data revealed that in pedagogical settings, the effective use of blogs and YouTube learning platforms enables knowledge sharing by connecting learners to contexts beyond the classroom. In turn, this facilitates the development of individual and critical voices and prompts individual accountability in learning. However, when asked to choose between study materials available on YouTube and those available on blogs or other social media platforms, students prefer the YouTube study materials over those available on blogs or other platforms.

Findings of their study further revealed that the students find the YouTube study material to be more user-friendly, which helps them retain the memory of the learning for a longer period of time. This is because of the audio-visual approach and easy availability of the contents on the site. Students who preferred blogs over YouTube content stated that academic blogging allows them to read and reflect in order to learn objectively driven study in more effective ways. It was noteworthy to see that the preference for YouTube or academic blogs did not appear to be significantly influenced by gender or age of the students.

15.7 SUMMARY

In this unit, we have discussed the various aspects of social media from a marketing research point of view. Organizations are constantly leveraging various marketing research techniques by using various social media platforms, which are efficient and cost effective as well. Social media provides and insight on their brand's goodwill, customer tastes and preferences, target market characteristics and other important facets required for market research. The role of social media in the marketing research process and other benefits of social media have also been discussed using a case study example of social media research insight.

15.8 KEY WORDS

Content Analysis: is a tool which is used to analyze the data collected in qualitative form.

Facebook Analytics: provides comprehensive, downloadable statistics on

your owned Pages

Social Listening: or social media listening refers to the process of identifying and monitoring the conversations or feedback about product, individual or organization from various social media channels.

Social Media: is characterized by user-generated content

Twitter Analytics: provides data on likes, responses, retweets, and other metrics to help you understand the reach and impact of Twitter ads.

15.9 SELF-ASSESSMENT QUESTIONS

1. Discuss the role of social media in marketing research
2. What are the benefits of using social media for marketing research?
3. Write down the tools of Social Media Marketing Research.
4. Describe the process of social media marketing research with example.
5. Write a brief note on how Content Analysis could be an effective data collection technique with suitable examples.

15.10 PROJECT QUESTION

A manufacturer of bakery items markets several brands of cookies to a chain of supermarkets. The product category is very competitive, and competitors use a great deal of newspaper and television advertising, with some moving into social media advertising. The owner of the firm desires more information for making promotional decisions on social media. Design a social media marketing research process for the firm.

15.11 FURTHER READINGS

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